

KNOWFUL[®] FILM COATED TABLET 800MG

Ingredient(s):

Each tablet contains :

Piracetam 800mg

Pharmacology (Summary of Pharmacodynamic and Pharmacokinetic) :

Pharmacodynamics:

1. Piracetam is a psychotropic agent which acts directly on the brain to improve the efficacy of the telencephalon in both normal subjects and those suffering from some functional deficit. This area of the brain is involved in cognition and also has a role to play in learning and memory, in alertness and in consciousness. Piracetam does not produce either sedation or stimulation.
2. Piracetam can act on the central nervous system in a variety of ways. It will modify neurotransmission within the brain, and can help to improve the metabolic environment essential for good neuronal function. It is also a haemorrhological agent and can improve microcirculation without producing vasodilation.
3. When given as acute or long-term treatment for patients suffering from a functional CNS deficit, it will heighten alertness and increase cognitive function. These changes are seen as a significant increase in the α - and β - activity, with a reduction in δ -activity, on an EEG trace.
4. Piracetam will protect and restore cognitive functional capacity after cerebral trauma, e.g. hypoxia or intoxication, and after electroshock therapy.
5. Piracetam may be given alone or together with other drugs when treating cortical myoclonia. It will reduce the duration of vestibular nystagmus.
6. Piracetam will also improve regional oxygen and glucose uptake in the brain in patients suffering from dementia subsequent to multiple infarcts, or in those with cerebral ischaemia.
7. Piracetam will inhibit the increased aggregation of activated platelets and, in conditions where there is abnormal rigidity of the red blood cell, it can restore deformability and the ability to pass through the microvasculature.

Pharmacokinetics:

1. When given by mouth, in either solute or tablet form, Piracetam is rapidly and almost totally absorbed in the gut. Bioavailability is almost 100%. When given as a single 2g dose, the peak blood level of 40-60 mg/mL is attained after 30 min. in the cerebrospinal fluid, the peak concentration is achieved at 2-8 hours post-dosage. The apparent volume of distribution is in the region of 0.6 L/kg. Plasma half life is 4-5 hours, while the half-life in the cerebrospinal fluid is 6-8 hours. The half-life is increased in cases of renal insufficiency.
2. Piracetam does not bind to plasma proteins and is eliminated unchanged principally via the kidney. Renal excretion is almost complete, i.e. >95%, after 30 hours. Renal clearance of Piracetam in healthy volunteers is 86 mL/min.
3. Piracetam will diffuse into all types of tissue and can cross both blood-brain barrier and the placenta, as well as the membranes employed in renal dialysis.
4. Piracetam is active in its unchanged form and is not metabolised by any of the animal species tested.
5. Piracetam is concentrated in the cerebral cortex, in the frontal, parietal and occipital lobes, in the cerebellum and in the basal ganglia.

Indication(s):

For elderly suffering from memory loss, vertigo, lack of concentration or of alertness, changes of mood, a deterioration in behaviour and personal negligence demonstrate an improvement in symptoms and also in the treatment of coma as well as cognitive deficit as a result of head injury, whether or not this is associated with vertigo.

Dosage & Administration:

The total daily dose is administered twice daily, but may also be given in 3 or 4 separate doses.

Cerebrovascular accident: If oral route permits, initially 12 g/day for 4 weeks, then 4.8 g/day for at least 12 weeks. Maintenance: 800mg 3 times a day. Long term therapy for psycho-organic syndrome in the elderly: 1.2-2.4 g/day.

Cognitive deficits resulting from head injury, whether or not these are associated with vertigo and also in the treatment of coma, the initial dose can vary from 9-12g daily. Maintenance therapy thereafter is a daily oral dose of 2.4g, given for a period of not less than 3 weeks.

To be taken orally.

Contraindication(s):

1. Patient who previously shown hypersensitivity to Piracetam.
2. Severe renal insufficiency, as demonstrated by a creatinine clearance of <20mL/min.

Warning and Precautions:

1. As the principal route of elimination for Piracetam is via the kidney, special care must be taken when treating patients known to suffer from renal insufficiency. Monitoring of renal function is recommended in such cases.
2. The increase in half-life is directly related to the decrease in renal function and creatinine clearance. This is also true for the elderly patients in whom creatinine clearance is dependent on age.

3. When the creatinine clearance is <60mL/min, or the serum creatinine is > 1.25 mg/100mL, the dosage prescribed should be calculated.

Creatinine Clearance (mL/min)	Serum Creatinine (mg/100mL)	Dosage
60-40	1.25-1.7	½ of normal dose
40-20	1.7-3	¼ of normal dose

Interaction with Other Medicaments:

1. The classical treatment for alcoholism, i.e. vitamins and sedatives, may continue to be prescribed during cures for alcoholism, in cases where there is a known vitamin deficiency or severe excitability.
2. Only 1 case of interaction has been reported, where Piracetam and thyroid extract (T3 and T4) were given together and confusion, irritability and sleep disturbances were later observed.
3. No other drug interactions have been described.

Pregnancy and Lactation:

Use in pregnancy & lactation:

Piracetam should not be prescribed during pregnancy or when breastfeeding, except under exceptional circumstances. Piracetam is able to cross the placenta.

Side Effects:

1. The side effects reported include nervousness, agitation, irritability, anxiety, and sleep disturbances. The incidence of these during clinical trials was $\leq 5\%$ and they were more often noted in the elderly patients taking >2.4g daily. In majority of the cases, a dose reduction suffices to make these symptoms disappear.
2. Some patients may complain of fatigue or drowsiness.
3. Gastrointestinal problems e.g. nausea, vomiting, diarrhoea and stomachache have also been reported, but their incidence during clinical trials was $\leq 2\%$.
4. Other symptoms, e.g. vertigo, headache, trembling and sexual stimulation have occasionally been reported.

Symptoms and Treatment of Overdose:

Piracetam appears to be devoid of toxicity even at very high doses and, therefore, the need for specific measures to be taken in case of an overdose is avoided.

However, the patient's general condition should be closely monitored. Close attention should be given to keep the patient well-hydrated and monitor the urine flow.

Storage condition:

Store at temperature below 30°C. Protect from light and moisture.

Shelf Life:

3 years from the date of manufacture.

Product Description:

An orange film coated tablet, one side is in the shape of “ ”



Dosage Forms and Packaging available:

Blister packing of 10's x 10.



Manufacturer and Product Registration Holder:
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